

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Electrical Engineering In-Semester Examination II Academic Year: 2025-26 (Semester – I)	
Class: TY B.Tech	Subject: Digital Signal Processing	Subject Code: PCC-EE 304
Date: 17.10.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed
- Non Programmable calculator is allowed.
- Figures to the right indicate full marks.
- The symbols carry the usual meaning.

Q.N.	Attempt The Following Questions	Marks	CO	BT
Q.1	Find discrete time fourier transform of the given signal $x(n) = \{ \underset{\uparrow}{1}, -1, 2, 2 \}$	4	CO2	BT1
Q.2	A causal discrete time LTI system is described by $y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n)$ Determine a) System Function b) Impulse Response	6	CO2	BT5
Q.3	Find IDFT of given sequence using IDFT formula. $X(K) = \{ 6, 0, -2, 0 \}$	4	CO2	BT1
Q.4	Compute the circular convolution using DFT and IDFT method for following sequences. $X1(n) = \{ 2, 1, 2, -1 \}$ and $X2(n) = \{ 1, 2, 3, 4 \}$	6	CO2	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	To understand benefits and limitations of processing signals digitally and properties of discrete-time LTI systems.
CO2	Represent and analyze the discrete-time signals and LTI systems in the frequency domain using Discrete-Time Fourier Transform (DTFT), z-transform and Discrete Fourier transform (DFT) tools.
CO3	To Implement DFT efficiently using Fast Fourier Transform (FFT) algorithms and use in practical applications.
CO4	Design an FIR or IIR filter for the specifications given in frequency domain
CO5	To realize digital system whose coefficients are known using hardware or software.
CO6	To propose and design a digital system for simple real application.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electrical Engineering
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (Semester I)

Class: Third Year B.Tech. Course: Elective: Electrical Machine Design Course Code: PEC-EE305

Date: 11.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

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Q. N.	Attempt all questions.	Marks	CO	BT
Q. 1	Illustrate the clean and clear diagrams of the laminations used in the stator and rotor of a large-size induction motor.	5	CO3	BT2
Q. 2	Explain the method to calculate the area of stator slots and diameter of stator core of an induction motor?	5	CO3	BT2
Q. 3	Calculate the main dimensions and turns per phase of a 250 hp, 3-phase, 50 Hz, 400 V, 1410 rpm slip ring induction motor. Assume $B_{av} = 0.5 \text{ Wb/m}^2$, $a_c = 30000 \text{ A/m}$, efficiency = 0.9 and power factor = 0.9, winding factor = 0.955, current density = 3.5 A/mm^2 . The ratio of core length to pole pitch is 1.2. The machine is delta connected.	5	CO3	BT3
Q. 4	Why flux density distribution is considered at 60° for calculation of magnetizing mmf? Explain this with suitable diagram.	5	CO4	BT1/ BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	Students will be able to:
CO 1	Explain the constructional details and design aspects of various types of transformers and calculate parameters such as EMF per turn, core dimensions, and insulation. (Understand, Apply)
CO 2	Evaluate transformer performance using equivalent circuit modeling, leakage reactance, and temperature rise analysis, and design cooling systems like tanks and radiators. (Evaluate, Design)
CO 3	Describe the construction and design process of three-phase induction motors including stator and rotor components, and calculate parameters like turns per phase, air gap length, and rotor currents.
CO 4	Analyze the performance of three-phase induction motors using circle diagram and derive key parameters such as maximum torque, power factor, and dispersion coefficient. (Analyze, Evaluate)
CO 5	Explain the construction and design methodology of synchronous machines and estimate main dimensions using electric/magnetic loadings and short-circuit ratio. (Understand, Apply)
CO 6	Apply computer-aided design techniques and optimization approaches to develop improved designs of transformers and induction motors. (Apply, Design)



Class: Third Year B.Tech.

Course Name: Signals and Systems

Course Code: PCC-EE305

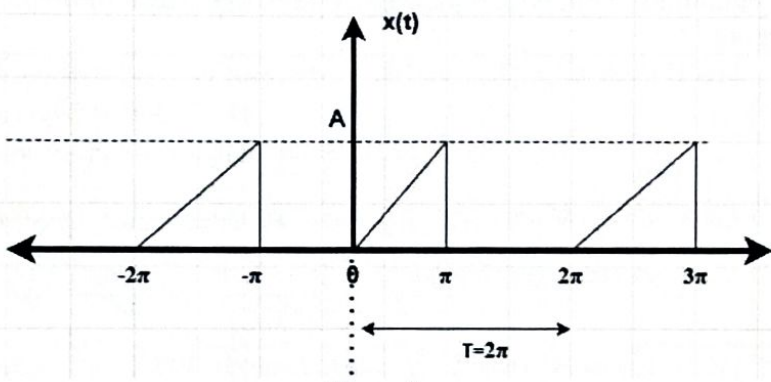
Date: 13.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

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Q.N.	Question Statement	Marks	CO	BT
Q.1	Calculate the trigonometric Fourier series coefficients, a_0 and a_n of the wave form shown in the figure 1.  Figure 1.	8	CO3	BT3
Q.2	For a real-valued, periodic signal $x(t)$ with period T. Recall (mention) the properties of the Fourier series corresponding to the following statement: i. If $x(t) \leftrightarrow C_n$ and $y(t) \leftrightarrow D_n$ then: $\alpha x(t) + \beta y(t) \leftrightarrow \alpha C_n + \beta D_n$ ii. If $x(t) \leftrightarrow C_n$ then: $x(-t) \leftrightarrow C_{-n}$	2	CO3	BT1
Q.3	Calculate the Fourier transform $x(j\omega)$, for the given exponential signal $x(t) = e^{-at}u(t)$; $a > 0$	5	CO4	BT3
Q.4	i. The Fourier transform of a function $x(t)$ is $X(j\omega)$. What will be the Fourier transform of $\frac{d x(t)}{dt}$? ii. What is the use of Fourier transform? iii. The Fourier transform of a function $x_1(t)$ is $X_1(j\omega)$ and for $x_2(t)$ is $X_2(j\omega)$. What will be the Fourier transform of $x_1(t) * x_2(t)$. iv. What is the Fourier Transform of a time-shifted signal $x(t-t_0)$? v. What is the effect of conjugation $x^*(t)$ on the Fourier Transform?	5	CO4	BT1

== ALL THE BEST ==



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Nanded**
Department of Electrical Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class: TY B.Tech	Subject: Microprocessor & Microcontroller	Subject Code: PCC-EE 303
Date: 13.10.2025	Time: 11:00 to 11:45	Max. Marks: 20

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Q.N.	Attempt The Following Questions	Marks	CO	BT
Q.1	Calculate the size of memory if starting location is 7000 H and last location is AFFF H	2	CO1	BT1
Q.2	Draw the timing diagram for MVI A, 05 H.	6	CO1	BT3
Q.3	Interface 8KB EPROM and 16 KB RAM with 8085 microprocessor using the chips 4KB of EPROM and 8 KB of RAM.	6	CO1	BT3
Q.4	Explain in detail programmable peripheral interface (PPI) (8255).	6	CO4	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 – Remember, BT2 – Understand, BT3 – Apply, BT4 – Analyze, BT5 – Evaluate, BT6 – Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	To describe basics of 8085, 8051 and its instruction set.
CO2	To understand historical development of microcontrollers and to know different 8, 16, 32 bit microcontrollers.
CO3	To solve assembly language programs based on the instruction set of 8085 and 8051.
CO4	To get insight of 8051 based hardware system and so to study ADC, keyboard etc.
CO5	To execute assembly language programs based on the instruction set of 8051
CO6	To develop 8085, 8051 based instrumentation system.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electrical Engineering
In-Semester Examination - II
Academic Year: 2025-26 (Semester – I)

Class: Third Year	Subject: Feedback Control System	Subject Code – PCC-EE303/302
Date: 10/10/2025	Time: 11:00am to 11:45am	Max. Marks: 20

Note:

- All questions are compulsory*
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Q.N.	Questions	Marks	CO	BT
Q.1	a) Define the following terms (i) delay time (ii) rise time (iii) peak time (iv) peak overshoot (v) settling time (vi) steady state error	5	CO-3	BT-1
	b) Find the steady state Error to :i) A unit step input ii) a unit ramp input iii) a unit parabolic input , consider the manginute of signal 'A' $G(s) = 10/s^2(s+4)(s^2+3s+12)$	5	CO-3	BT-1
Q.2	a) Find the range of values of 'k' so that system with following characteristics equation will be stable: $f(s) = s(s^2+5s+1)(s+4)+k=0$ by using Routh's Method.	5	CO-4	BT-1
	b) Explain Routh's stability Criterion and with anyone example.	5	CO-4	BT-2



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electrical Engineering
In-Semester Examination - II
Academic Year: 2025-26 (Semester – I)

Class: Third Year	Subject: Power System Engineering	Subject Code – PCC-EE301
Date: 09/10/2025	Time: 11:00am to 11:45am	Max. Marks: 20

Note:

- i. All questions are compulsory
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Q.N.	Questions	Marks	CO	BT
Q.1	a) Explain the nominal- π model of a medium transmission line with its equivalent circuit and find the ABCD parameters.	4	CO-3	BT-2
	b) An overhead 3-phase transmission line delivers 5000 kW at 22 kV and 0.8 power factor lagging. The resistance and reactance of each conductor are 4Ω and 6Ω respectively. Solve for: (i) the sending end voltage, (ii) the percentage voltage regulation.	6	CO-3	BT-3
Q.2	a) Derive the sending-end voltage and current equations of a medium line using the Nominal-T method and illustrate with phasor diagram.	4	CO-3	BT-2
	b) Neglecting resistance and use a base of 100 MVA and 220 kV in 50Ω line. Solve for the per-unit reactances and draw the per-unit impedance diagram for the given power system. The ratings of the generator, motor and transformers are Generator : 30 MVA, 25 kV, $X = 20\%$ Motor : 40 MVA, 11 kV, $X = 30\%$ Y- Y Transformer: 40 MVA, 33/220 kV, $X = 15\%$ Y- Δ Transformer: 30 MVA, 220/11 kV, $X = 15\%$	6	CO-4	BT-3